# Reading Notes: Analytic Combinatorics

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# Chapter 1

# COMBINATORIAL STRUCTURES AND ORDINARY GENERATING FUNCTIONS

## 1.1 Symbolic enumeration methods

▷ **I.2.** *Unimodal permutations.* A unimodal permutation can be viewed as a sequence of elements increasing to a peak and then decreasing. This structure is represented by:

$$\mathcal{P}^{\mathcal{U}} = \textsc{Set}(\mathcal{Z}) \star \mathcal{Z}^{\blacksquare} \star \textsc{Set}(\mathcal{Z})$$

$$P^U(z) = \int_0^z (e^t \cdot \frac{d}{dt} t \cdot e^t) dt = \frac{1}{2}(e^{2z} - 1)$$

which yields

$$P_n^U = n![z^n]\frac{1}{2}e^{2z} = n!\frac{1}{2}\frac{2^n}{n!} = 2^{n-1}$$

Alternatively,

$$P_n^U = \sum_{k=1}^{n} \binom{n-1}{k-1} = \sum_{j=0}^{n-1} \binom{n-1}{j} = 2^{n-1}$$

where $k$ is the position of $n$ in the permutation.

## 1.2 Admissible constructions and specifications

## 1.3 Integer compositions and partitions

▷ **I.16.** *Partitions with summands bounded in number and size.* The number of partitions of size $n$ with at most $k$ summands, each at most $l$, is

$$[z^n]\frac{(1-z)(1-z^2)\cdots(1-z^{k+l})}{((1-z)(1-z^2)\cdots(1-z^k))\cdot((1-z)(1-z^2)\cdots(1-z^l))} \tag{1.1}$$

Let $f(k,l)$ denote the generating function where the coefficient of $z^n$ represents the number of such partitions. To derive a functional equation, consider the value of the largest

part. There are two mutually exclusive possibilities: the largest part is strictly less than $l$, or it is exactly $l$. This distinction yields:

$$f(k,l) = f(k,l-1) + z^l f(k-1,l) \tag{1.2}$$

We can derive 1.1 from this relation by introducing $F_k(x)$, the generating function for $f(k,l)$ with respect to the index $l$:

$$F_k(x) = \sum_{l=0}^{\infty} f(k,l)x^l$$

For the base case $k = 0$, the only valid partition is the empty partition of size 0, implying $f(0,l) = 1$ for all $l \geq 0$. Thus:

$$F_0(x) = \sum_{l=0}^{\infty} 1 \cdot x^l = \frac{1}{1-x}$$

Summing 1.2 over $l \geq 1$ after multiplying by $x^l$, we obtain:

$$\sum_{l=1}^{\infty} f(k,l)x^l - \sum_{l=1}^{\infty} f(k,l-1)x^l = \sum_{l=1}^{\infty} f(k-1,l)(zx)^l$$

- The first term simplifies to $F_k(x) - 1$, since $f(k,0) = 1$.
- The second term is simply $xF_k(x)$.
- The right-hand side yields $F_{k-1}(zx) - 1$.

Plugging these in:

$$(1-x)F_k(x) - 1 = F_{k-1}(zx) - 1 \implies F_k(x) = \frac{1}{1-x}F_{k-1}(zx)$$

Now we apply this functional relation repeatedly:

$$\begin{aligned} F_k(x) &= \frac{1}{1-x}F_{k-1}(zx) \\ &= \frac{1}{1-x}\frac{1}{1-zx}F_{k-2}(z^2x) \\ &= \frac{1}{(1-x)(1-zx)\cdots(1-z^{k-1}x)}F_0(z^kx) \end{aligned}$$

Since $F_0(x) = \frac{1}{1-x}$, we have $F_0(z^kx) = \frac{1}{1-z^kx}$. Thus:

$$F_k(x) = \sum_{l=0}^{\infty} f(k,l)x^l = \frac{1}{(1-x)(1-zx)\cdots(1-z^kx)}$$

Since all the roots of the denominator are distinct (assuming $|z| < 1$ or that $z$ is a formal variable), we can decompose this into a sum of partial fractions:

$$\frac{1}{\prod_{i=0}^{k}(1-z^ix)} = \sum_{j=0}^{k} \frac{A_j}{1-z^jx}$$

To find a specific $A_j$, we multiply both sides by $(1 - z^j x)$ and then set $x = z^{-j}$:

$$\begin{aligned} A_j &= \left[\frac{1 - z^j x}{\prod_{i=0}^{k}(1 - z^i x)}\right]_{x=z^{-j}} \\ &= \frac{1}{\prod_{i=0}^{j-1}(1 - z^i z^{-j}) \prod_{i=j+1}^{k}(1 - z^i z^{-j})} \\ &= \frac{(-1)^j z^{j(j+1)/2}}{\prod_{i=1}^{j}(1 - z^i) \prod_{i=1}^{k-j}(1 - z^i)} \end{aligned}$$

The coefficient of $x^l$ in $F_k(x)$ is the sum of the coefficients of $x^l$ in each partial fraction. For a term $\frac{A_j}{1-z^j x}$, the coefficient is $A_j (z^j)^l$.

$$\begin{aligned} f(k,l) &= \sum_{j=0}^{k} A_j (z^j)^l \\ &= \sum_{j=0}^{k} \frac{(-1)^j z^{j(j+1)/2} \cdot z^{jl}}{\prod_{i=1}^{j}(1 - z^i) \prod_{i=1}^{k-j}(1 - z^i)} \\ &= \frac{1}{\prod_{i=1}^{k}(1 - z^i)} \sum_{j=0}^{k} \binom{k}{j}_z (-1)^j z^{j(j+1)/2 + jl} \end{aligned}$$

where $\binom{k}{j}_z$ is the Gaussian binomial coefficient:

$$\begin{aligned} \binom{k}{j}_z &= \frac{\prod_{i=1}^{k}(1 - z^i)}{\prod_{i=1}^{j}(1 - z^i) \prod_{i=1}^{k-j}(1 - z^i)} \\ &= \frac{(1 - z^k)(1 - z^{k-1}) \cdots (1 - z^{k-j+1})}{(1 - z)(1 - z^2) \cdots (1 - z^j)} \end{aligned}$$

We can rewrite the exponent in the sum:

$$\frac{j(j+1)}{2} + jl = \frac{j(j-1)}{2} + j(l+1)$$

Now the sum becomes

$$f(k,l) = \frac{1}{\prod_{i=1}^{k}(1 - z^i)} \sum_{j=0}^{k} \binom{k}{j}_z (-1)^j z^{j(j-1)/2} (z^{l+1})^j$$

The $q$-binomial theorem states that for a variable $X$:

$$\prod_{i=0}^{k-1}(1 - z^i X) = \sum_{j=0}^{k} \binom{k}{j}_z (-1)^j z^{j(j-1)/2} X^j \tag{1.3}$$

Plugging $X = z^{l+1}$ into the theorem yields 1.1:

$$\begin{aligned} f(k,l) &= \frac{(1 - z^{l+1})(1 - z^{l+2}) \cdots (1 - z^{l+k})}{(1 - z^1)(1 - z^2) \cdots (1 - z^k)} \\ &= \frac{(1 - z)(1 - z^2) \cdots (1 - z^{k+l})}{((1 - z)(1 - z^2) \cdots (1 - z^k)) \cdot ((1 - z)(1 - z^2) \cdots (1 - z^l))} \end{aligned}$$

To prove 1.3, let $P_k(X)$ be the product we are expanding:

$$P_k(X) = \prod_{i=0}^{k-1}(1 - z^i X)$$

We want to show that $P_k(X) = \sum_{j=0}^{k} \binom{k}{j}_z (-1)^j z^{j(j-1)/2} X^j$. Consider the relationship between $P_k(X)$ and $P_{k-1}(X)$. By definition:

$$P_k(X) = (1 - z^{k-1}X)P_{k-1}(X)$$

Now notice what happens if we replace $X$ with $zX$ in $P_k(X)$:

$$P_k(zX) = \prod_{i=0}^{k-1}(1 - z^{i+1}X) = \frac{1 - z^k X}{1 - X} P_k(X)$$

Rearranging this gives us a functional equation for $P_k(X)$:

$$(1 - X)P_k(zX) = (1 - z^k X)P_k(X)$$

Let $P_k(X) = \sum_{j=0}^{k} c_j X^j$. We substitute this power series into the functional equation:

$$(1 - X)\sum_{j=0}^{k} c_j (zX)^j = (1 - z^k X)\sum_{j=0}^{k} c_j X^j$$

Now, we compare the coefficients of $X^j$ on both sides. For a specific power $j$:

$$c_j z^j - c_{j-1} z^{j-1} = c_j - c_{j-1} z^k$$

Rearranging to solve for $c_j$ in terms of $c_{j-1}$:

$$c_j = c_{j-1}\frac{z^{j-1} - z^k}{z^j - 1} = -c_{j-1} z^{j-1}\frac{1 - z^{k-j+1}}{1 - z^j}$$

Starting from the base case $c_0 = P_k(0) = 1$, we can find $c_j$ by iteration:

$$c_1 = -c_0 z^0 \frac{1 - z^k}{1 - z^1}, \quad c_2 = -c_1 z^1 \frac{1 - z^{k-1}}{1 - z^2} = (-1)^2 z^{0+1}\frac{(1 - z^k)(1 - z^{k-1})}{(1 - z^1)(1 - z^2)}$$

The exponent of $z$ becomes $0 + 1 + 2 + \cdots + (j-1) = \frac{j(j-1)}{2}$. The fraction is the definition of the Gaussian binomial coefficient $\binom{k}{j}_z$. Thus, the coefficient is:

$$c_j = (-1)^j z^{j(j-1)/2}\binom{k}{j}_z$$

Plugging the coefficients back into the sum, we obtain:

$$\prod_{i=0}^{k-1}(1 - z^i X) = \sum_{j=0}^{k}\binom{k}{j}_z (-1)^j z^{j(j-1)/2} X^j$$

This completes the proof.

▷ **I.17.** *Systems of linear Diophantine inequalities.*

$$\{x_1 \geq 0,\ x_2 \geq 2x_1,\ x_3 \geq 2x_2,\ x_4 \geq 2x_3\}$$

We introduce non-negative slack variables $y_j \in \mathbb{Z}_{\geq 0}$ to quantify the "excess" of each term relative to its lower bound.

$$x_1 = y_1 \tag{1.4}$$
$$x_2 = 2x_1 + y_2 = 2y_1 + y_2 \tag{1.5}$$
$$x_3 = 2x_2 + y_3 = 4y_1 + 2y_2 + y_3 \tag{1.6}$$
$$x_4 = 2x_3 + y_4 = 8y_1 + 4y_2 + 2y_3 + y_4 \tag{1.7}$$

The total weight is $S = x_1 + x_2 + x_3 + x_4 = 15y_1 + 7y_2 + 3y_3 + y_4$. Since $y_j$ are independent and range freely over non-negative integers, the OGF $F(z)$ is the product of the individual generating functions for each $y_j$:

- For $15y_1$ : $\sum_{y_1=0}^{\infty}(z^{15})^{y_1} = \frac{1}{1-z^{15}}$
- For $7y_2$ : $\sum_{y_2=0}^{\infty}(z^{7})^{y_2} = \frac{1}{1-z^{7}}$
- For $3y_3$ : $\sum_{y_3=0}^{\infty}(z^{3})^{y_3} = \frac{1}{1-z^{3}}$
- For $y_4$ : $\sum_{y_4=0}^{\infty}(z^{1})^{y_4} = \frac{1}{1-z^{1}}$

Multiplying these together yields the desired OGF:

$$F(z) = \frac{1}{(1-z)(1-z^3)(1-z^7)(1-z^{15})}$$

$$\{x_1 + x_2 \leq x_3\}$$

Following the same approach as in the previous problem, we use slack variables to convert the inequality into an equality of independent parts. The constraint $x_3 \geq x_1 + x_2$ can be rewritten by introducing a non-negative integer $y \geq 0$:

$$x_3 = x_1 + x_2 + y$$

The size of the composition is $S = x_1 + x_2 + x_3 + x_4 = 2x_1 + 2x_2 + y + x_4$. Since the variables $x_1, x_2, y, x_4$ are independent and each ranges freely over the non-negative integers, the OGF is the product of the generating functions for the individual terms in the sum:

- For $2x_1$ : $\frac{1}{1-z^2}$
- For $2x_2$ : $\frac{1}{1-z^2}$
- For $y$ : $\frac{1}{1-z}$
- For $x_4$ : $\frac{1}{1-z}$

Multiplying these together gives:

$$F(z) = \frac{1}{(1-z^2)^2(1-z)^2}$$

$$\{x_1 + x_2 \geq x_3\}$$

Since the inequality does not immediately suggest a single slack variable, a more robust approach uses the complement method. Let $U(z) = (1-z)^{-4}$ be the OGF for four unrestricted variables. The universe of compositions partitions into those where $x_1+x_2 < x_3$ and our target class $x_1 + x_2 \geq x_3$. We find the OGF for the complement $x_1 + x_2 < x_3$. Set $x_3 = x_1 + x_2 + 1 + y$, where $y \geq 0$. The total sum $S$ is:

$$S = x_1 + x_2 + (x_1 + x_2 + 1 + y) + x_4 = 2x_1 + 2x_2 + y + x_4 + 1$$

The OGF for this complement, which we denote by $C(z)$, is:

$$C(z) = \frac{z}{(1-z^2)^2(1-z)^2}$$

The OGF for our constraint is then:

$$F(z) = U(z) - C(z) = \frac{1}{(1-z)^4} - \frac{z}{(1-z^2)^2(1-z)^2} = \frac{1+z+z^2}{(1-z)^4(1+z)^2}$$

$$\{x_1 + x_2 \leq x_3 + x_4\}$$

To find the OGF for $\{x_1 + x_2 \leq x_3 + x_4\}$, we can use the complement method combined with the symmetry of the variables. Let $A = x_1 + x_2$ and $B = x_3 + x_4$. We seek the OGF for compositions satisfying $A \leq B$. In the universe of all possible compositions $(x_1, x_2, x_3, x_4)$, there are three mutually exclusive cases: 1. $A < B$, 2. $A > B$, 3. $A = B$. By symmetry, the OGF for case 1 must be identical to the OGF for case 2, since $A$ and $B$ are built from the same number of variables with the same weights. Denote this common OGF by $G_<(z)$, and let $G_=(z)$ be the OGF for case 3. The total unrestricted OGF is $U(z) = \frac{1}{(1-z)^4}$. Therefore:

$$U(z) = 2G_<(z) + G_=(z)$$

Our target OGF is $F(z) = G_<(z) + G_=(z)$. Solving the system gives:

$$F(z) = \frac{U(z) + G_=(z)}{2}$$

Now $G_=(z)$ is the OGF for compositions in which the sum of the first two variables equals the sum of the last two.

- The number of compositions with $x_1 + x_2 = n$ is $[z^n]\frac{1}{(1-z)^2} = n + 1$.
- The number of compositions with $x_3 + x_4 = n$ is the same.
- For the equality $x_1 + x_2 = x_3 + x_4$ to hold, both sums must equal some $n \geq 0$.

$$G_=(z) = \sum_{n=0}^{\infty}(n+1)^2 z^{2n} = \frac{1+z^2}{(1-z^2)^3}$$

Substituting $U(z)$ and $G_=(z)$ into our formula yields:

$$F(z) = \frac{1}{2}\left[\frac{1}{(1-z)^4} + \frac{1+z^2}{(1-z^2)^3}\right] = \frac{1+z+2z^2}{(1-z)^4(1+z)^3}$$

$$\{x_1 \leq x_2,\ x_2 \geq x_3,\ x_3 \leq x_4\}$$

Because the inequalities alternate direction (up, down, up), we cannot use a straightforward chain of slack variable substitutions as we did for $x_1 \leq x_2 \leq x_3 \leq x_4$. Instead, we apply the inclusion–exclusion principle to the constraints. We begin with the universe of all unrestricted compositions, whose OGF is $U(z) = \frac{1}{(1-z)^4}$. We wish to enforce three conditions:

- $C_1 : x_1 \leq x_2$
- $C_2 : x_2 \geq x_3$
- $C_3 : x_3 \leq x_4$

However, it is simpler to work with chains of "less than or equal" inequalities. Consider the complement of the downward middle step. Take the set of all compositions satisfying both $x_1 \leq x_2$ and $x_3 \leq x_4$. From this set, we subtract those compositions that violate the middle constraint (i.e., $x_2 < x_3$).
The OGF for the set defined by $x_1 \leq x_2$ and $x_3 \leq x_4$: These two pairs are independent.

- For $x_1 \leq x_2$, set $x_2 = x_1 + y_1$. The total contribution is $2x_1 + y_1$, so the OGF is $\frac{1}{(1-z^2)(1-z)}$.
- For $x_3 \leq x_4$, the OGF is identically $\frac{1}{(1-z^2)(1-z)}$.

Since the two pairs are independent, we multiply their OGFs:

$$A(z) = \frac{1}{(1-z^2)^2(1-z)^2}$$

Now we find the OGF for the violation $x_1 \leq x_2 < x_3 \leq x_4$ by introducing slack variables:

- $x_1 = y_1$
- $x_2 = y_1 + y_2$
- $x_3 = y_1 + y_2 + 1 + y_3$
- $x_4 = y_1 + y_2 + 1 + y_3 + y_4$

Summing the four expressions gives:

$$S = 4y_1 + 3y_2 + 2y_3 + y_4 + 2$$

The OGF for this violation is therefore:

$$V(z) = \frac{z^2}{(1-z^4)(1-z^3)(1-z^2)(1-z)}$$

The OGF for the zigzag constraint is the OGF for the two independent pairs minus the OGF for the cases where the peak at $x_2$ fails:

$$\begin{aligned} F(z) &= \frac{1}{(1-z^2)^2(1-z)^2} - \frac{z^2}{(1-z^4)(1-z^3)(1-z^2)(1-z)} \\ &= \frac{1}{(1-z)(1-z^2)}\left[\frac{1}{(1-z)(1-z^2)} - \frac{z^2}{(1-z^3)(1-z^4)}\right] \end{aligned}$$

▷ **I.18.** *Odd versus distinct summands.* We establish a one-to-one correspondence between partitions into odd parts and partitions into distinct parts.

- Odd → Distinct
  1. Group identical odd parts together.
  2. If an odd part $k$ appears $m$ times, write $m$ in binary: $m = 2^{a_1} + 2^{a_2} + \cdots$ with distinct powers of two.
  3. Replace the $m$ copies of $k$ by $2^{a_1} \cdot k,\ 2^{a_2} \cdot k,\ \ldots$
- Distinct → Odd
  1. Factor each part as $2^a \times$ odd.
  2. Split it into $2^a$ copies of that odd factor.
  3. Do this for all parts; now all parts are odd.
  4. Sort the resulting parts to obtain a partition into odd parts.

As an example:

$$\begin{aligned} 3+3+3+3+3+1+1 &= 5 \times 3 + 2 \times 1 \\ &= (2^2 + 2^0) \times 3 + 2^1 \times 1 \\ &= 4 \times 3 + 1 \times 3 + 2 \times 1 \\ &= 12 + 3 + 2 \end{aligned}$$

$$\begin{aligned} 12+3+2 &= 4 \times 3 + 3 + 2 \times 1 \\ &= (3+3+3+3) + 3 + (1+1) \\ &= 3+3+3+3+3+1+1 \end{aligned}$$

▷ **I.19.** *Euler's pentagonal number theorem.* To find a recurrence for the partition numbers, we consider the reciprocal of the generating function $P(z)$:

$$A(z) = \prod_{n=1}^{\infty}(1 - z^n)$$

Euler's Pentagonal Number Theorem states that this product expands into the following sparse power series:

$$\begin{aligned} A(z) &= \sum_{k=-\infty}^{\infty} (-1)^k z^{k(3k+1)/2} \\ &= \sum_{k=-\infty}^{\infty} (-1)^k z^{k(3k-1)/2} \\ &= 1 - z^1 - z^2 + z^5 + z^7 - z^{12} - z^{15} + \cdots \end{aligned}$$

The exponents $g_k = \frac{k(3k-1)}{2}$ are the generalized pentagonal numbers. The product of $P(z)$ and $A(z)$ must equal 1:

$$\sum_{n=0}^{\infty} p(n)z^n \cdot \sum_{k=-\infty}^{\infty} (-1)^k z^{g_k} = 1$$

Comparing the coefficient of $z^n$ on both sides yields the algorithmically useful recurrence:

$$p(n) = p(n-1) + p(n-2) - p(n-5) - p(n-7) + p(n-12) + p(n-15) \cdots$$

A Python implementation is provided below:

```python
def compute_partitions(N):
P = [0] * (N + 1)
P[0] = 1  # Base case: p(0) = 1

for i in range(1, N + 1):
    k = 1
    while True:
        # Calculate pentagonal numbers for k and -k
        g_k_pos = k * (3 * k - 1) // 2
        g_k_neg = k * (3 * k + 1) // 2

        # Determine sign: +1 for k=1,2 (terms 1,2), -1 for k=3,4 (terms 3,4)...
        # Simplest way: (-1)^(k-1)
        sign = 1 if k % 2 == 1 else -1

        # Add p(i - g_k_pos)
        if g_k_pos <= i:
            P[i] += sign * P[i - g_k_pos]
        else:
            break # No more terms will fit

        # Add p(i - g_k_neg)
        if g_k_neg <= i:
            P[i] += sign * P[i - g_k_neg]
        else:
            break

        k += 1
return P
```

▷ **I.20.** *A digital surprise.*

$$\begin{aligned}
\varphi &= (1-10^{-1})(1-10^{-2})(1-10^{-3})(1-10^{-4})\cdots \\
&= \left[\prod_{n=1}^{\infty}(1-z^n)\right]_{z=.1} \\
&= \left[1 - z^1 - z^2 + z^5 + z^7 - z^{12} - z^{15} + z^{22} + z^{26} - \cdots\right]_{z=.1} \\
&= .\dot{9} - .1 - .01 + .00001 + .0000001 - .000000000001 - \cdots
\end{aligned}$$

The partial sums are:

- $\varphi_1 = .999999999999999999999999$
- $\varphi_2 = .899999999999999999999999$
- $\varphi_3 = .889999999999999999999999$
- $\varphi_4 = .890009999999999999999999$
- $\varphi_5 = .890010099999999999999999$

- $\varphi_6 = .8900100999989999999999999$
- $\varphi_7 = .8900100999989989999999999$
- $\varphi_8 = .8900100999989990000000999$

▷ **I.21.** *Lattice points.* We are interested in the number of integer solutions to the inequality $\sum_{i=1}^{d} x_i^2 \leq n^2$. First, consider a single coordinate $x_i \in \mathbb{Z}$. The "size" contributed by this coordinate to the sum of squares is $x_i^2$. The generating function for the possible values of a single coordinate, where the exponent of $z$ represents the square of the coordinate, is:

$$\Theta(z) = \sum_{x \in \mathbb{Z}} z^{x^2} = 1 + 2\sum_{n=1}^{\infty} z^{n^2}$$

Since each coordinate $x_1, \cdot, x_d$ is chosen independently, the generating function for the sum of squares $\sum x_i^2$ is the $d$-th power of the individual generating function:

$$S(z) = (\Theta(z))^d = \sum_{k=0}^{\infty} r_d(k) z^k$$

In this series, the coefficient $[z^k]S(z)$ (denoted as $r_d(k)$) is exactly the number of ways to represent the integer $k$ as a sum of $d$ squares. Geometrically, this is the number of lattice points on the sphere (the boundary) of radius $\sqrt{k}$. The number of points $N(n, d)$ in the closed ball of radius $n$ is:

$$N(n, d) = \sum_{k=0}^{n^2} r_d(k)$$

In generating functions, the prefix sum of a sequence's coefficients is obtained by multiplying the generating function by $\frac{1}{1-z}$. Therefore:

$$\sum_{k=0}^{\infty} \left( \sum_{j=0}^{k} r_d(j) \right) z^k = \frac{1}{1-z} S(z) = \frac{1}{1-z} (\Theta(z))^d$$

To find the total count up to $k = n^2$, we simply extract the coefficient of $z^{n^2}$:

$$N(n, d) = [z^{n^2}] \frac{1}{1-z} (\Theta(z))^d$$

## 1.4 Words and regular languages

▷ **I.23.** *Alice, Bob, and coding bounds.* Following Proposition I.4, the class $\mathcal{C}$ of binary strings avoiding the pattern 11 has the OGF:

$$C(z) = \frac{1+z}{1-z-z^2}$$

To account for a variable-length coding scheme where codewords are selected from $\bigcup_{j=0}^{L} \mathcal{C}_j$, we consider the sum of the first $L+1$ coefficients. This is represented by the OGF:

$$S(z) = \frac{C(z)}{1-z} = \frac{1+z}{(1-z)(1-z-z^2)}$$

The asymptotic growth of the sum $\sum_{j=0}^{L} C_j$ is governed by the smallest singularity of $S(z)$. Since the pole at $z = 1/\varphi$ (where $\varphi = \frac{1+\sqrt{5}}{2}$) is closer to the origin than the pole at $z = 1$, the exponential growth is dominated by $C(z)$. Specifically, as $L \to \infty$:

$$\sum_{j=0}^{L} C_j \sim A \cdot \varphi^L$$

for some constant $A$. To find the required code length $L(n)$ for $n$ bits of information, we solve for $2^n \leq \sum_{j=0}^{L} C_j$ by taking the base-2 logarithm:

$$n \leq \log_2 \left( \sum_{j=0}^{L} C_j \right) = L \log_2 \varphi + O(1)$$

Rearranging for $L$, we obtain the lower bound:

$$L \geq \frac{n}{\log_2 \varphi} + O(1) \approx 1.440420n + O(1)$$

▷ **I.27.** *Congruence languages.* The DFA $\mathcal{A}$ for binary strings congruent to 2 (mod 7) is structured as follows:

- **States**: A set $\mathcal{Q} = \{S_0, \ldots, S_6\}$, where each state $S_r$ represents the current remainder $r$ modulo 7.
- **Initial State**: $S_0$ (representing the value 0).
- **Accepting State**: $\overline{Q} = \{S_2\}$.
- **Alphabet**: $\mathcal{A} = \{0, 1\}$.

Each transition follows the rule $q_r \circ b = q_{(2r+b) \pmod 7}$:

Table 1.1: State Transition

| Current State (Remainder) | Input 0 ($2r \pmod 7$) | Input 1 ($2r + 1 \pmod 7$) |
|---|---|---|
| $S_0$ | $S_0$ | $S_1$ |
| $S_1$ | $S_2$ | $S_3$ |
| $S_2$ | $S_4$ | $S_5$ |
| $S_3$ | $S_6$ | $S_0$ |
| $S_4$ | $S_1$ | $S_2$ |
| $S_5$ | $S_3$ | $S_4$ |
| $S_6$ | $S_5$ | $S_6$ |

▷ **I.29.** *The forward equations.* By a "last-letter decomposition," the set $\mathcal{M}_k$ of words leading to state $q_k$ from the initial state $q_0$ satisfies the combinatorial relation:

$$\mathcal{M}_k \cong [\![k = 0]\!] \cdot \{\epsilon\} + \sum_{j,\alpha : q_j \circ \alpha = q_k} (\mathcal{M}_j \times \{\alpha\})$$

where $[\![\cdot]\!]$ is the Iverson bracket. This translates to a linear system of OGFs:

$$M_k(z) = \delta_{k,0} + z \sum_j M_j(z) T_{j,k}$$

In vector notation, with $\mathbf{M}(z)$ as a column vector and $\mathbf{e}_0$ the first basis vector, the system is a transposed version of the backward equations:

$$\mathbf{M}(z)^T = \mathbf{e}_0^T + z\mathbf{M}(z)^T T$$

## 1.5 Tree structures

▷ **I.42.** *Faster determination of Schröder numbers.*

$$S(z) = \frac{1}{4}(1 + z - \sqrt{1 - 6z + z^2})$$

Consider in general

$$p(z)S(z) + q(z) = \sqrt[k]{r(z)}$$

where $p(z)$, $q(z)$, and $r(z)$ are functions of $z$. Taking the derivative with respect to $z$,

$$p'(z)S(z) + p(z)S'(z) + q'(z) = \frac{1}{k}r(z)^{\frac{1}{k}-1}r'(z)$$

Multiplying both sides by $r(z)$,

$$p'(z)r(z)S(z) + p(z)r(z)S'(z) + q'(z)r(z) = \frac{1}{k}(p(z)S(z) + q(z))r'(z)$$

Rearranging,

$$p(z)r(z)S'(z) + (p'(z)r(z) - \frac{1}{k}p(z)r'(z))S(z) = \frac{1}{k}q(z)r'(z) - q'(z)r(z)$$

Plugging $p(z) = -4$, $q(z) = 1 + z$, $k = 2$, and $r(z) = 1 - 6z + z^2$ into the equation and simplifying yields

$$(z^2 - 6z + 1)S'(z) + (3 - z)S(z) = 1 - z$$

from which the recurrence follows.

## 1.6 Additional constructions

## 1.7 Perspective

# Chapter 2

# LABELLED STRUCTURES AND EXPONENTIAL GENERATING FUNCTIONS

## 2.1 Labelled classes

## 2.2 Admissible labelled constructions

## 2.3 Surjections, set partitions, and words

## 2.4 Alignments, permutations, and related structures

## 2.5 Labelled trees, mappings, and graphs

## 2.6 Additional constructions

## 2.7 Perspective

# Chapter 3

# COMBINATORIAL PARAMETERS AND MULTIVARIATE GENERATING FUNCTIONS

Content for chapter 3 goes here...

## 3.1 An introduction to bivariate generating functions (BGFs)

## 3.2 Bivariate generating functions and probability distributions

## 3.3 Inherited parameters and ordinary MGFs

## 3.4 Inherited parameters and exponential MGFs

## 3.5 Recursive parameters

## 3.6 Complete generating functions and discrete models

## 3.7 Additional constructions

## 3.8 Extremal parameters

## 3.9 Perspective